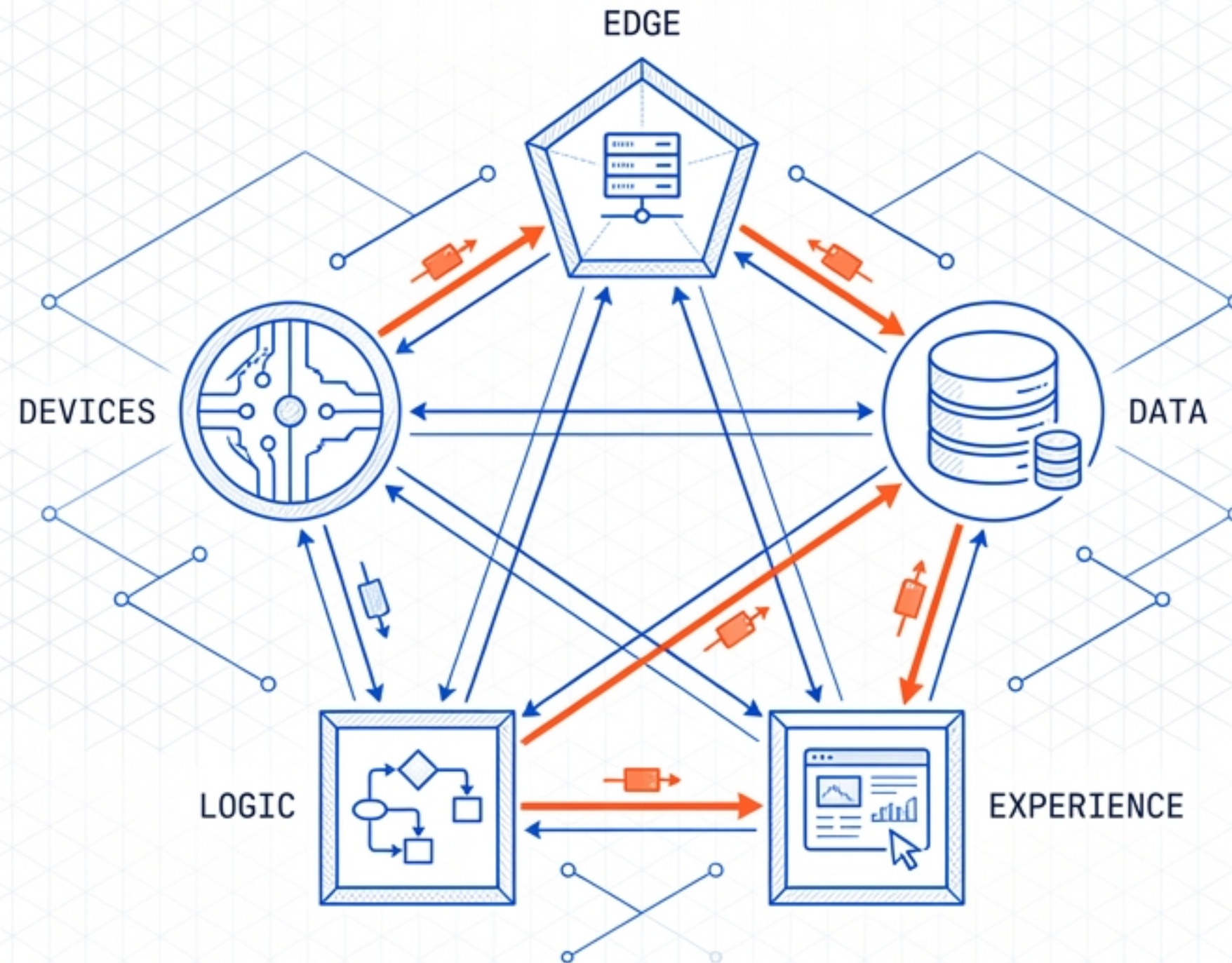


Building Scalable IoT: The Architect's Blueprint

A comprehensive technical guide to the Losant Enterprise IoT Platform.



Designed for Developers, Architects, and Product Owners

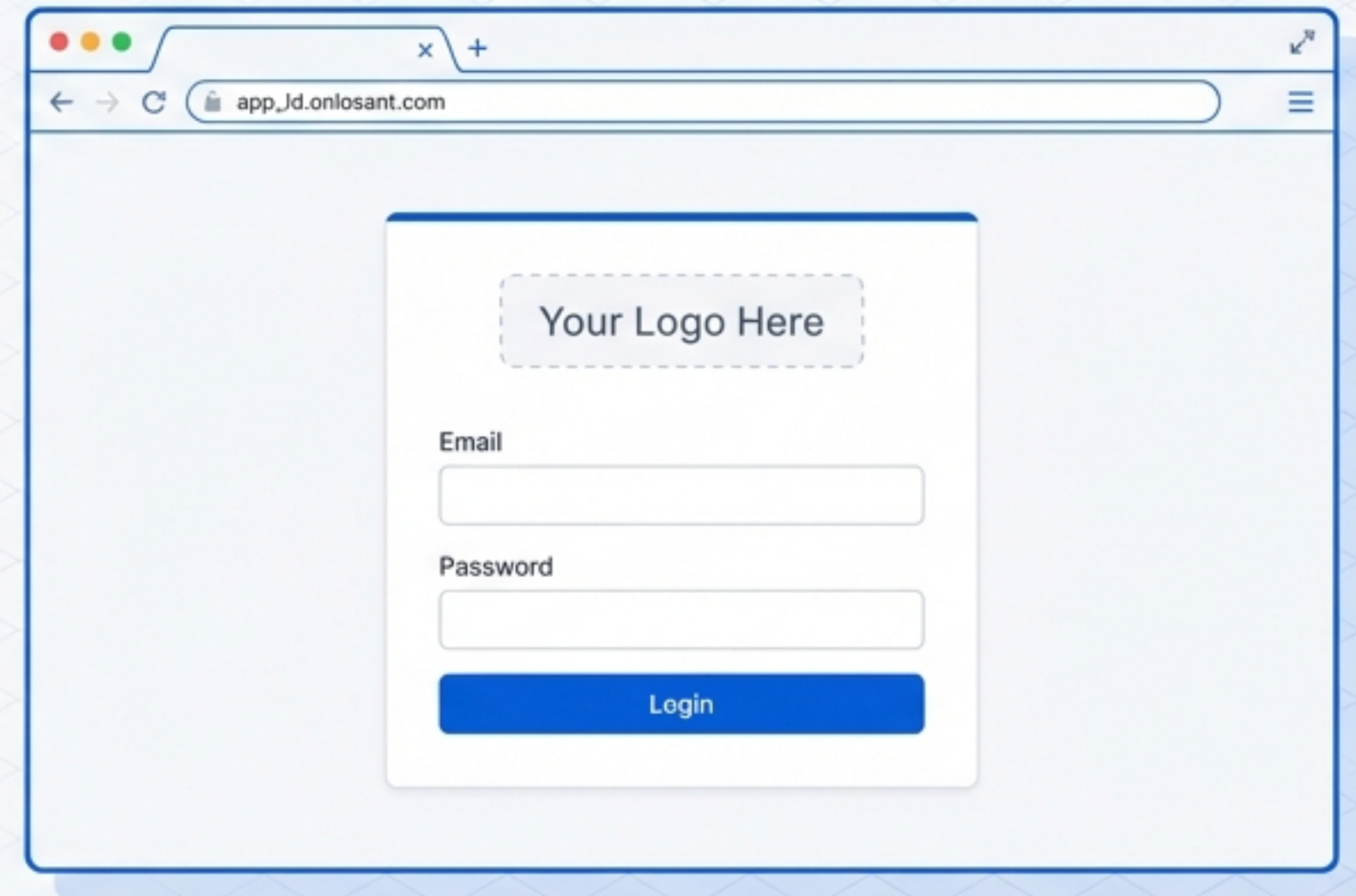
The Five Pillars of the Losant Ecosystem



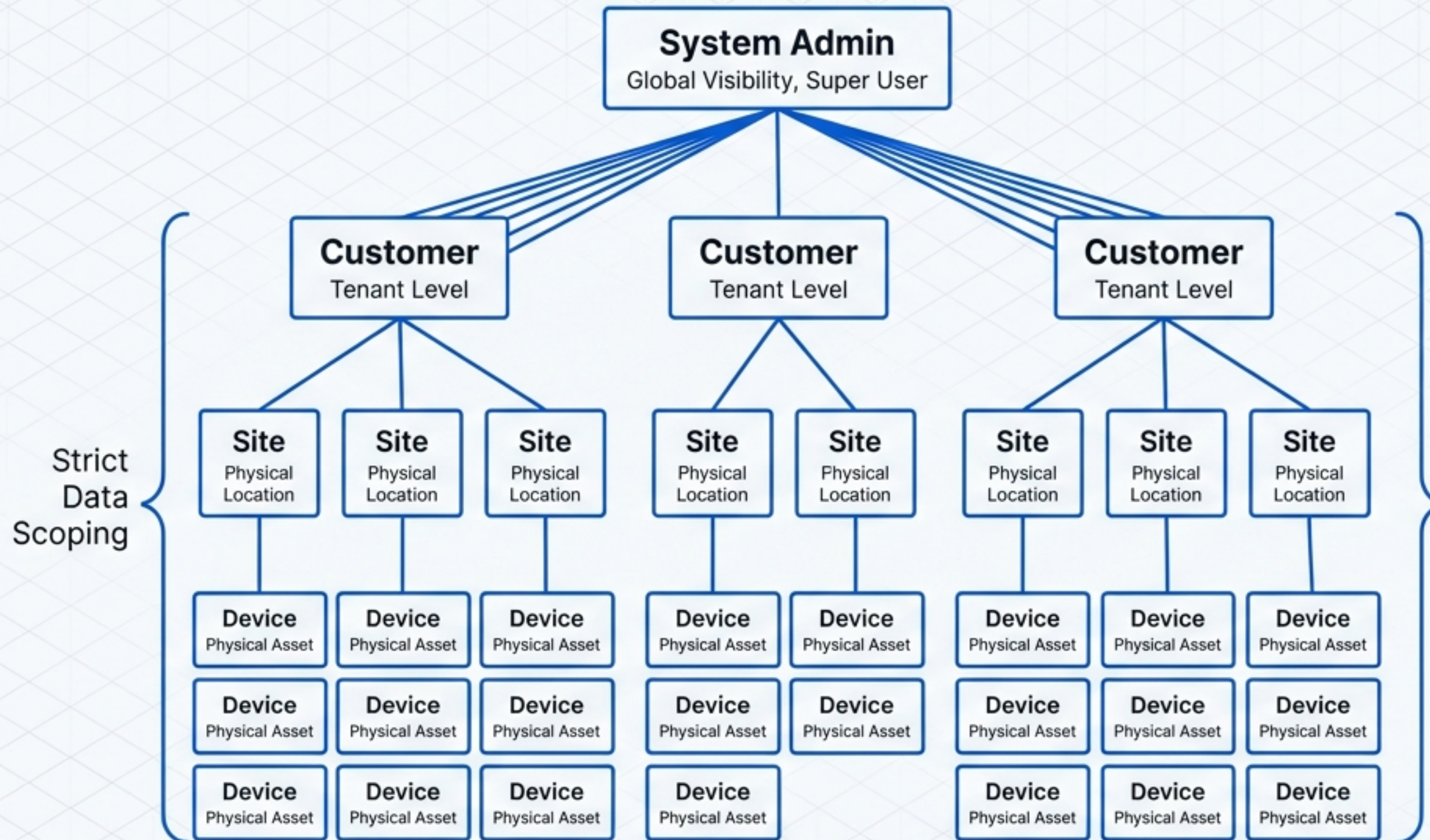
Accelerating Development with the Connected Product Foundation (CPF)

A production-ready application template, not just a feature set.

- ✓ **Multitenancy:** Native support for Customers, Sites, and Devices.
- ✓ **User Management:** Built-in invitations, password resets, and role-based access.
- ✓ **Device Management:** Registration workflows and device recipes.
- ✓ **White Labeling:** No-code customization of logos, fonts, and colors.



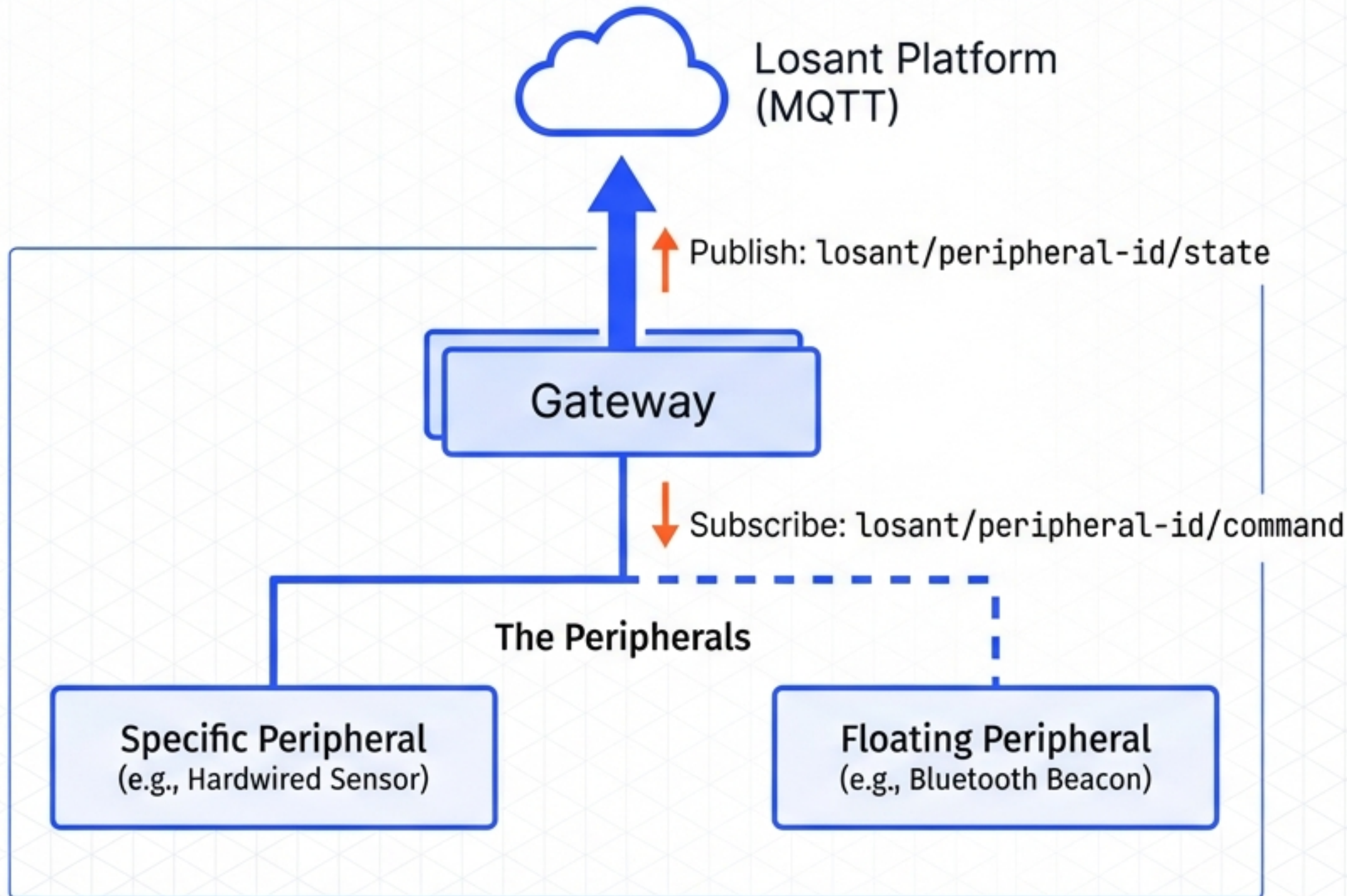
CPF Architecture: Hierarchy & Administration



User Roles & Permissions

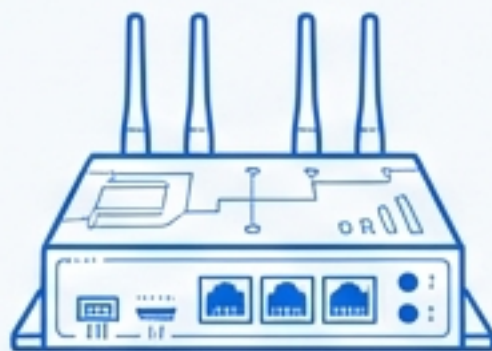
- **Viewer:** Read-only access to dashboards & events.
- **Editor:** Can modify device properties & handle events.
- **Admin:** Full control: Add/Remove Users, Devices, & Sites.

Device Connectivity: Gateways & Peripherals



Gateways report state on behalf of peripherals that cannot connect directly to the internet.

Edge Compute: Gateway Agent vs. Embedded Agent



Gateway Edge Agent (GEA)

Tech Stack:

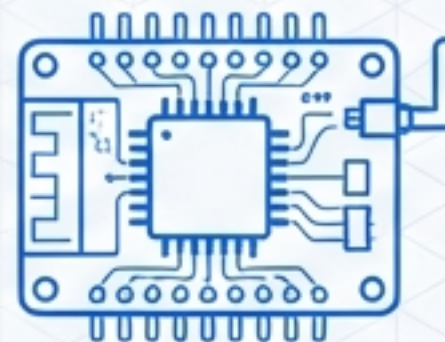
- Docker / Linux

Key Features:

- Local Broker
- Built-in Drivers (Modbus, BACnet, OPC UA)

Use Case:

Heavy industrial aggregation.



Embedded Edge Agent (EEA)

Tech Stack:

- WASM (WebAssembly) / C++

Key Features:

- No OS required
- Low Power
- Manual MQTT Client

Use Case:

Mass-deployed sensors & constrained hardware.

Advanced Integration: Custom Topics & Streaming

Section 1: Custom MQTT



Bypass standard topics for custom payloads.

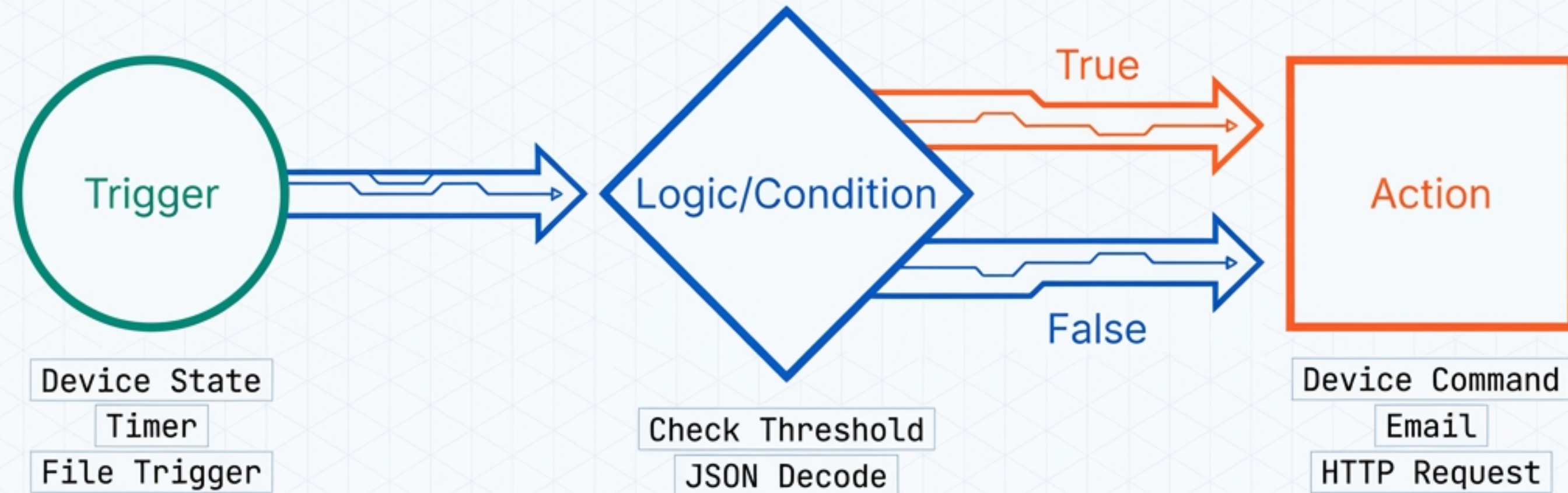
Section 2: Server-Sent Events



```
const evtSource = new EventSource('/api/stream');  
evtSource.addEventListener('mqttMessage', (e) => { ... });
```

Real-time push to browser without polling.

The Visual Workflow Engine: Application Logic



Execution Scopes

- Application (Cloud)
- Edge (Gateway)
- Embedded (Device)

Data Strategy: Time-Series vs. Data Tables

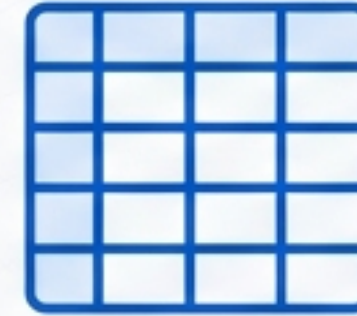


Time-Series Data

Device State

- High Volume
- Telemetry (GPS, Temp)
- Native Aggregations (Mean, Max)

Storage: Retention Policies apply.



Data Tables

Application Data

- Static/Metadata
- Fault Codes & User Lists
- Queryable (SQL-like)

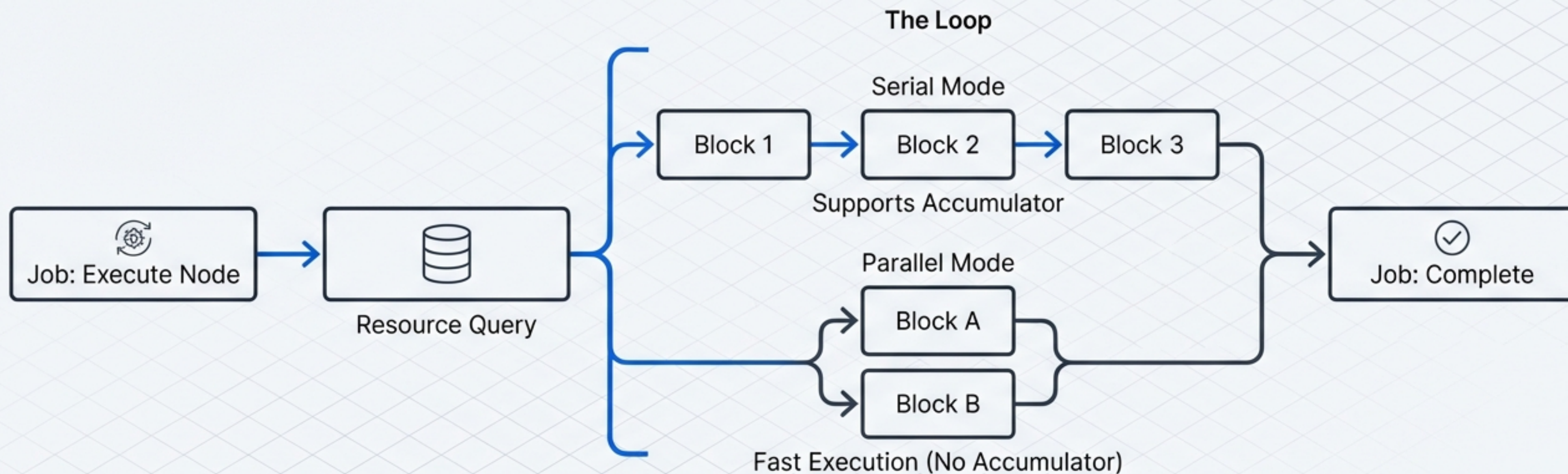
Constraints:

Unique

Required

Optional

Batch Processing: Resource Jobs



Key Workflow Nodes

- Job: Iteration Trigger
- ✓/✗ Job: Acknowledge Node (Success/Failure)

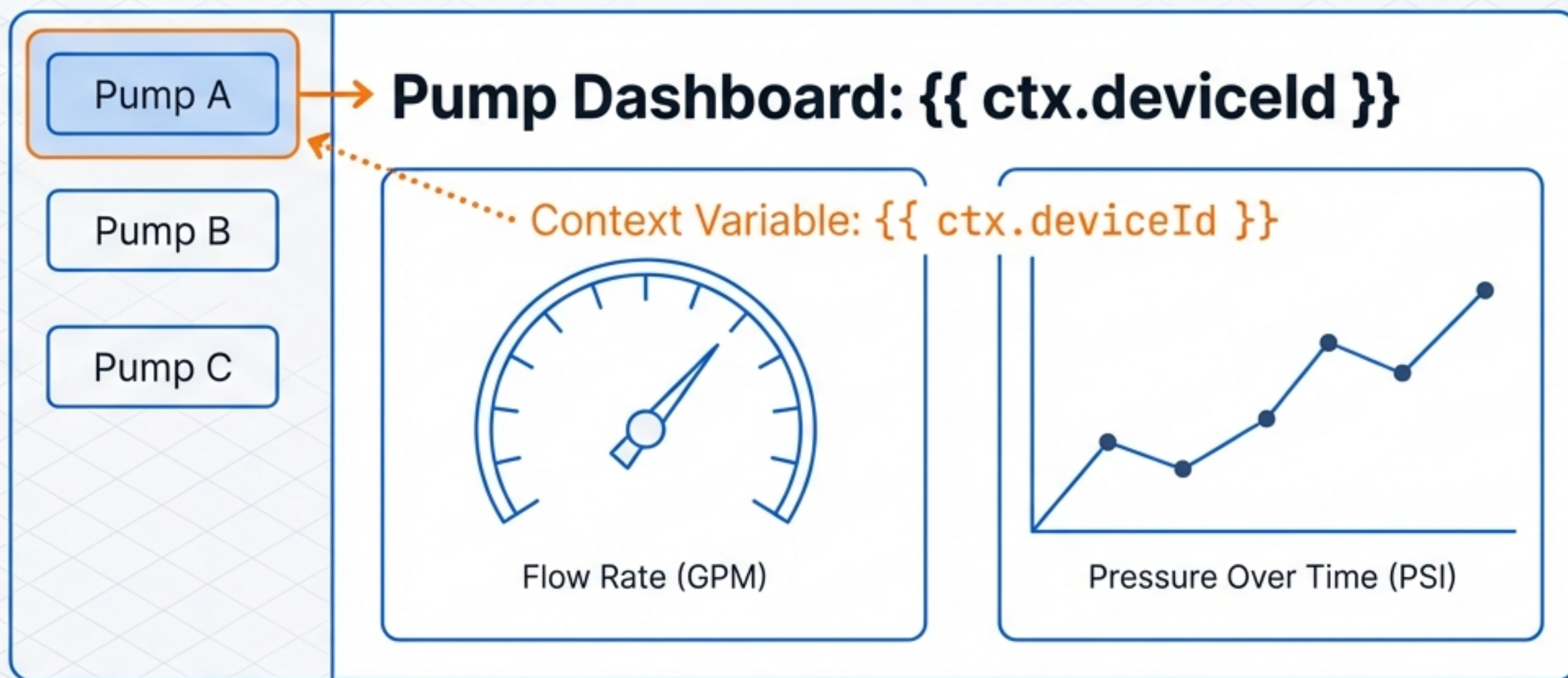
Deep Analytics: Jupyter Notebook Integration



```
import pandas as pd
import os
# Read Input
df = pd.read_csv(os.environ['INPUT_DIR'], 'data.csv')
# Process relative to Query Time
```

Data snapshots are relative to 'Query Time' for reproducible analysis.

Visualization Strategy: Context & Templates

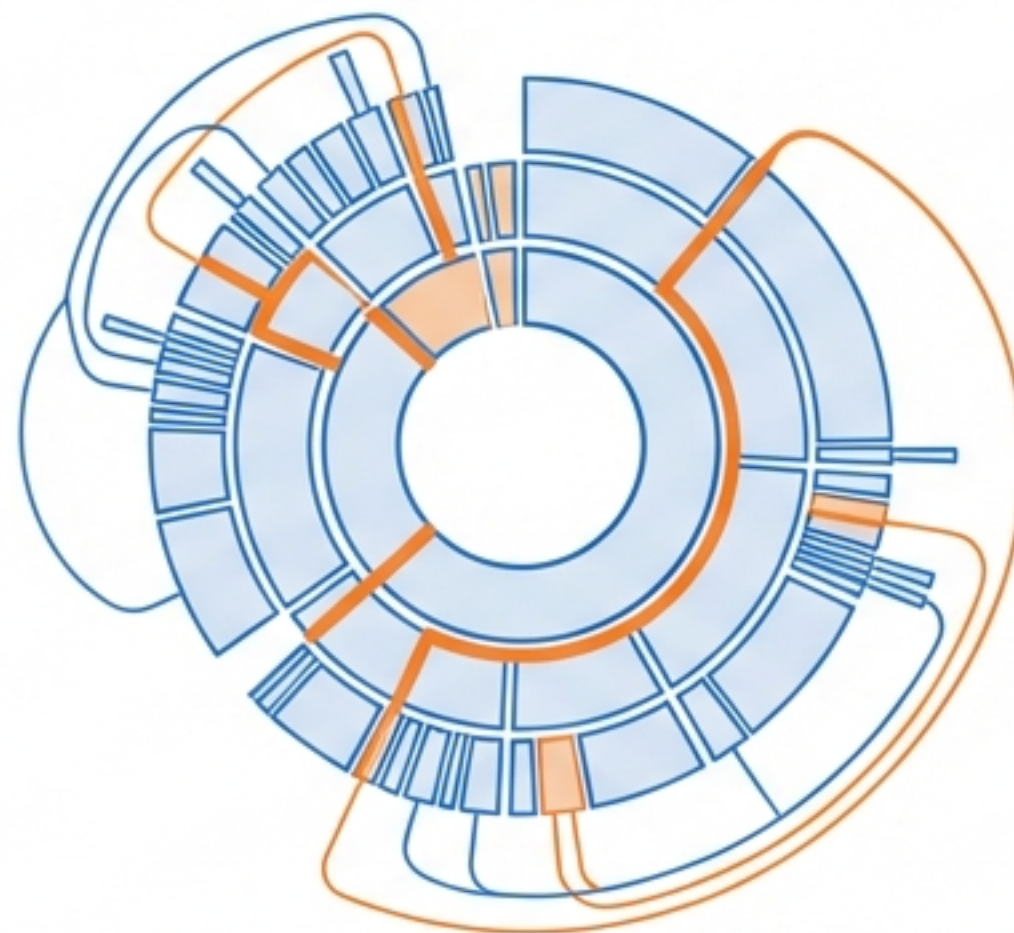


- **One Dashboard, Many Devices:** Define context variables to dynamically load data.
- **CPF Integration:** Device Recipes automatically link devices to their dashboard templates.

Advanced UI: The Custom HTML Block

Custom HTML / JS / CSS

```
DashboardBlock.on('queryChange', (e) => {  
  const val = e.queries.myQuery.value;  
  renderChart(val);  
});  
  
Dashboard.emitCustomEvent({ ... });
```



Sandboxed Iframe



Access to 3rd Party
Libraries (Chart.js, D3)



Cross-Block Communication
via `emitCustomEvent`

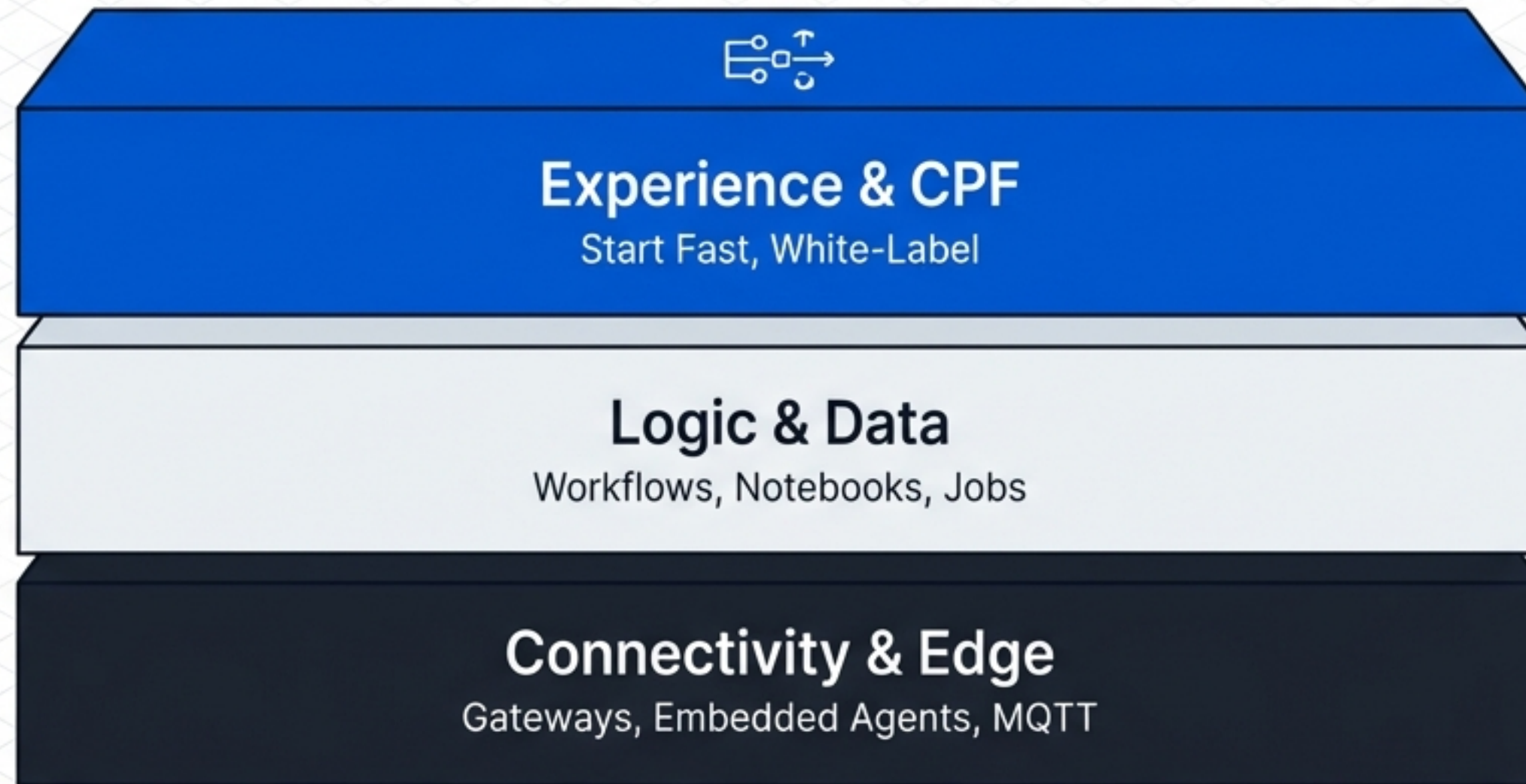
SCADA Visualization: Image Overlays & Logic



```
{{#gt data.temperature 100}}  
  {{colorMarker '#ff9900'}}  
{{else}}  
  {{colorMarker '#3333ff'}}  
{{/gt}}
```

Map digital data to physical spaces. Use conditional logic to change visual states dynamically.

Summary: From Blueprint to Production



Losant provides the blocks; you provide the architecture.